Approach — Team "Sanidhya and Aarichsun"

1. Understanding the problem

The challenge is a stationary 10-arm Bernoulli bandit with a strict 5,000-impression horizon. The core difficulty lies in the incredibly low-yield, densely clustered click-through rates (CTRs) of the top ads—averaging around 7% to 8%. Because the optimal ads are mathematically so close, standard algorithms either fail to distinguish between them or waste too much of the fixed budget endlessly double-checking their runner-up choices. We needed a policy that finds the winner quickly and completely shuts down exploration before the budget runs out.

2. What we chose and why

We ultimately implemented a **Horizon-Decaying UCB1** policy. Standard UCB1 calculates an upper confidence bound to guarantee we explore uncertain ads, but in a 7% CTR environment, its default math is way too aggressive and second-guesses itself constantly. To fix this, we introduced a dynamic exploration multiplier (c) that shrinks linearly as we run out of impressions: c = 0.55 \times (1.0 - t/5000).

- 'Early on (t is small), the multiplier is high, allowing rigorous testing to quickly eliminate the bottom-performing ads.'
- 'As t grows and we approach the 5,000 limit, c decays toward zero. This forces the algorithm into pure exploitation exactly when the value of learning new information vanishes.'

We chose this smooth decay over a "hard cutoff" because a hard cutoff is a gamble; if the best ad hits a string of bad luck exactly when exploration stops, the algorithm permanently locks onto the wrong choice.

3. What else we tried

Variant	**Leaderboard Score**	**Notes**
Standard Thompson Sampling	~376.6	Over-explores. The Beta distribution probability curves overlap too much for the closely clustered top ads.
Standard UCB1	~336.6	Plummeted. The strict uncertainty bonus is far too aggressive in a ~7% low-reward environment.

TS with Hard Horizon Cutoff	~385.8	Stronger, but vulnerable if the true best ad hits a bad luck streak right at the cutoff point.
Decaying Epsilon-Greedy	~388.8	Better late-game exploitation, but pure random exploration early on is less efficient than UCB.
Horizon-Decaying UCB1 (Submitted)	391.6	Best balance. Adapts risk tolerance perfectly to the remaining runway.

4. Results

- Local practice score: ~400+ clicks
- Best website leaderboard score: **391.6**, rank **#4**.

5. What we'd do next

We would explore tuning the decay curve of our c parameter—perhaps using an exponential decay rather than a linear one—to see if dropping exploration slightly faster in the mid-game yields better results. Alternatively, we would look into combining the logic of UCB with Thompson Sampling by implementing highly specific Informative Priors (setting the baseline Beta distribution to exactly match the 7.5% environment average) to ground the algorithm's early assumptions in reality.